Zisterzienser Ziffern mit LaTeX

samcarter

Dante Frühjahrstagung Lahnau

11. – 14. März 2026

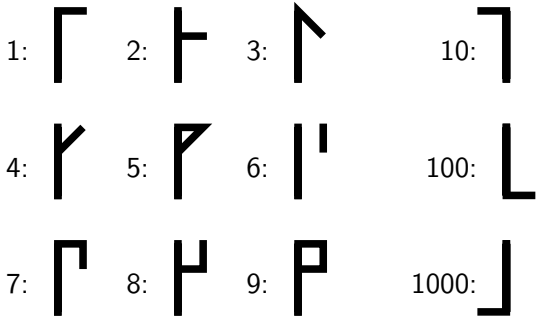
Titelbild: Michael Pabst / CC BY-SA 3.0 (via Wikimedia Commons)

Zisterzienser Ziffern

- mittelalterliches Zahlensystem
- Zisterzienser Orden, frühes 13. bis 15. Jhd
- Davor System von John of Basingstoke
- andere Verwendungen bis ins frühe 18. Jhd, z.B. Weinhandel um Brügge, Freimaurer in Paris
- Inspiration für andere Codes

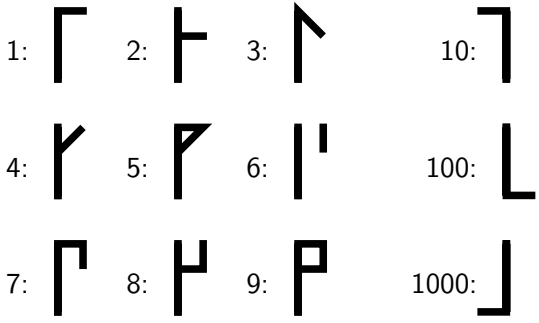
Liber Floridus (Ghent)

Zisterzienser Ziffern

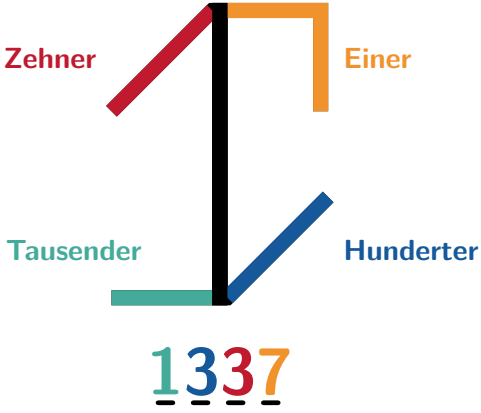


(nach einem Astrolabium aus dem frühen 16. Jhd)

Zisterzienser Ziffern

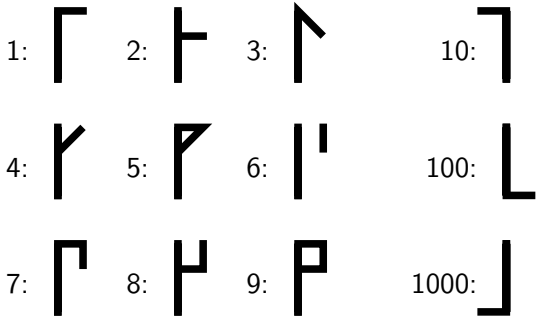


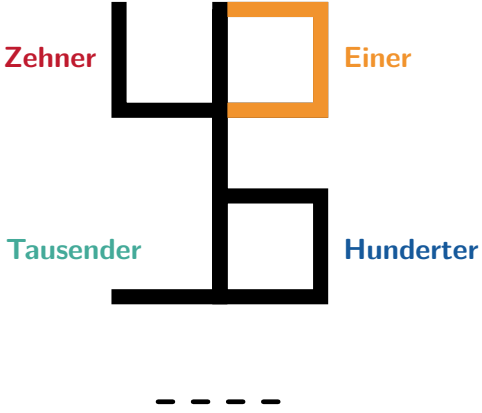
(nach einem Astrolabium aus dem frühen 16. Jhd)



(MCCCXXXVII)

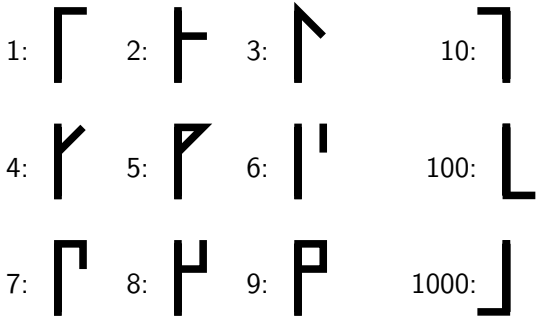
Zisterzienser Ziffern

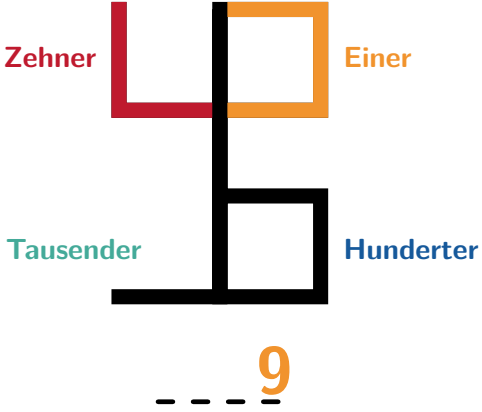




(nach einem Astrolabium aus dem frühen 16. Jhd)

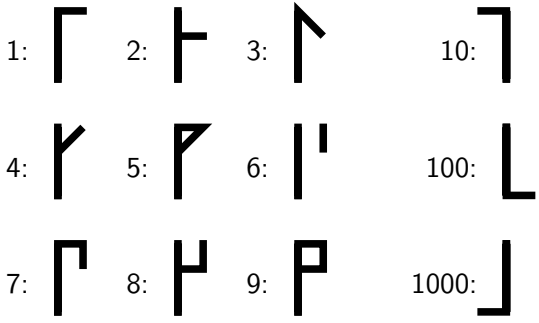
Zisterzienser Ziffern

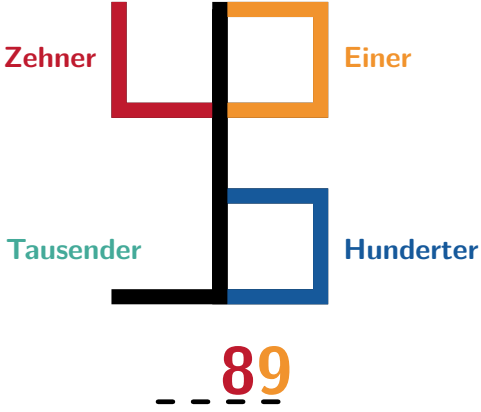




(nach einem Astrolabium aus dem frühen 16. Jhd)

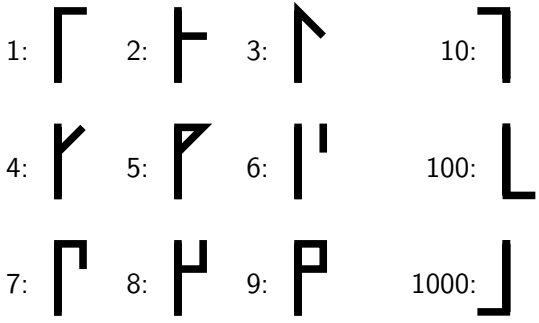
Zisterzienser Ziffern

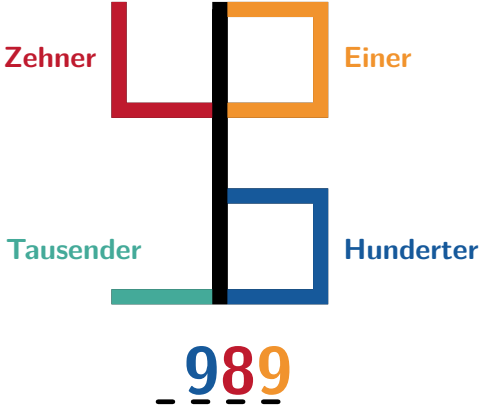




(nach einem Astrolabium aus dem frühen 16. Jhd)

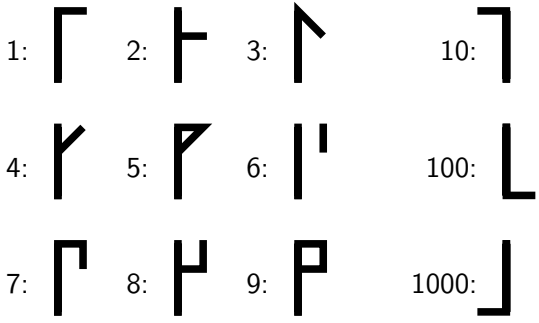
Zisterzienser Ziffern

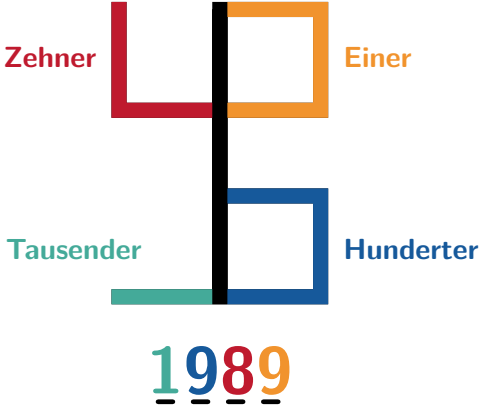




(nach einem Astrolabium aus dem frühen 16. Jhd)

Zisterzienser Ziffern





(nach einem Astrolabium aus dem frühen 16. Jhd)

LaTeX Pakete

Xistercian

- Autor: Jonathan Spratte
- sehr schnell
- viele Optionen
- Querformat und alt. Formen

Cistercian

- Autor: samcarter
- TikZ-basiert
- Linien-Übergänge
- Anpassung an Schriftart

Xistercian: Einfaches Beispiel

```latex
\documentclass{article}

\usepackage{xistercian}
\pagenumbering{cistercian}

\begin{document}

Ocean's \cistercian{page}

Die \cisterciannum{3} Musketiere

\end{document}
```

Ocean's T

Die ᛚ Musketiere

Xistercian: Anpassungsfähig

```latex
\color{red}\cisterciannum{101}
Dalmatiner
```

[Dalmatiner

```latex
\Huge Die \cisterciannum{2} Türme
```

Die ⊢ Türme

Xistercian: Optionen

```
\cisterciansetup{horizontal,redraw}
Reise um die Erde in
\cisterciannum{80} Tagen
```

Reise um die Erde in Tagen

```
\cisterciannum{99} Luftballons
\cisterciansetup{
  alternate-9=triangle,redraw}\par
\cisterciannum{99} Bottles of Beer
```

 Luftballons
 Bottles of Beer

```
\cisterciansetup{redraw,width=1em,
  height=2em,strokes=2pt}
Die \cisterciannum{4} Federn
```

Die Federn

Cistercian: Einfaches Beispiel

```latex
\documentclass{article}

\usepackage{cistercian}
\pagenumbering{cistercian}

\begin{document}

\cistercian{3} Haselnüsse für
Aschenbrödel

\end{document}
```

3 Haselnüsse für Aschenbrödel

Cistercian: Anpassungsfähig

```latex
\color{red} Jim Knopf und die wilde
\cistercian{13}
```

Jim Knopf und die wilde

```latex
\Huge Die \cistercian{7} Zwerge
```

Die Zwerge

```latex
\Huge\fontfamily{ugq}\selectfont
\cistercian{1001} Nacht
```

Nacht

Cistercian: Optionen

```
\cistercianset{stroke=1mm}

Ali Baba und die
\cistercian[height=1cm]{40} Räuber
```

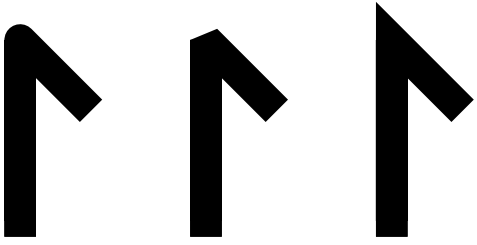
Ali Baba und die Räuber

Cistercian: TikZ-basiert

```
Goldlöckchen und die
\cistercian[draw=red]{3} Bären
```

Goldlöckchen und die Bären

```
\cistercianset{scale=8}
\cistercian[line join=round]{3}
\cistercian[line join=bevel]{3}
\cistercian[line join=miter]{3}
```



Panda: CleverPanda

```
\usepackage{CleverPanda}

\fontfamily{cmr}\selectfont
Computer Modern Roman:
\rule{ \CleverPanda{} }{1.6ex}

\fontfamily{ugq}\selectfont
URW Grotesk:
\rule{ \CleverPanda{} }{1.6ex}
```

Computer Modern Roman: |

URW Grotesk: |

Panda: BusyPanda

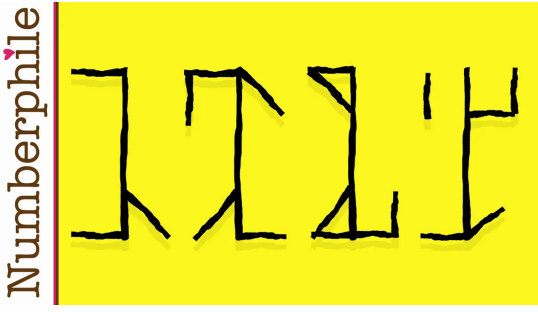
```
\usepackage{BusyPanda}

\fontfamily{cmr}\selectfont
Computer Modern Roman:
\BusyPanda{}

\fontfamily{ugq}\selectfont
URW Grotesk:
\BusyPanda{}
```

Computer Modern Roman: 1

URW Grotesk: 2.40491

Neugierig?

- Numberphile (2020). *The Forgotten Number System*. YouTube video. URL: `https://www.youtube.com/watch?v=9p55Qgt7Ciw`
- David A. King (2001). *The Ciphers of the Monks: A Forgotten Number-notation of the Middle Ages*. Stuttgart: Franz Steiner Verlag



Dieser Vortrag wurde ihnen präsentiert mit ltx-talk und talkthemespectrum ☺